Indian Statistical Institute
M. Tech. (CS) I
Machine Learning 1
Back Paper Examination, Date: June 12, 2024

Timing: 3 hrs. 15 mins — Maximum marks: 100

Answer any five questions. Use of a calculator is permitted.

1. Consider a dataset (x_i, y_i) for $i = 1, 2, \ldots, n$. The goal is to fit a linear regression model $y = \beta_0 + \beta_1 x$ using the Newton-Raphson method to minimize the sum of squared residuals:

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

- Derive the gradient vector and the Hessian matrix for the objective function $S(\beta_0, \beta_1)$.
- Derive the Newton-Raphson update steps for β_0 and β_1.
- Given the initial guesses $\beta_0^{(0)} = 0$ and $\beta_1^{(0)} = 0$, perform one iteration of the Newton-Raphson method. Use the following dataset:

x	y
1	2
2	3
3	5
4	7

(6+4+10 = 20)

2. Consider a dataset $\{x_1, x_2, \ldots, x_n\}$ consisting of n independent and identically distributed (i.i.d.) observations from an exponential distribution with an unknown rate parameter λ. The probability density function (PDF) of the exponential distribution is given by $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise.

- Derive the Maximum Likelihood Estimate (MLE) for the rate parameter λ.
- Assume a Gamma prior on λ with shape parameter α and rate parameter β, where the density function of λ is given by

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

  for $\lambda > 0$. Derive the Maximum A Posteriori (MAP) estimate for λ.
- Compare the MLE and MAP estimates and discuss the effect of the prior in the MAP estimate. What happens when the number of samples n becomes very large?

(6+8+6 = 20)

3. Consider a two-layer neural network with one input layer, one hidden layer with one neuron, and one output layer with one neuron. Assume a ReLU activation function for the hidden layer and a sigmoid activation function for the output layer. The network is trained using the backpropagation algorithm with a mean squared error loss function. The weights of the network are initialized as follows:

- Initial weights for the connections from the input layer to the hidden layer: $W_i^{(1)} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$
- Initial weight for the connection from the hidden layer to the output layer: $W^{(2)} = 0.3$

The input data and corresponding targets are as follows:

Input	Target
$(1,1)$	0.6
$(2,2)$	0.8

(a) Perform one forward pass of the input data through the neural network, computing the activations at the hidden layer and the output layer.

(b) Compute the total error (mean squared error) for the current predictions.

(c) Perform one backward pass (backpropagation) to compute the gradients of the weights.

(d) Update the weights using gradient descent with a learning rate of 0.1.

(6+4+6+4 = 20)

4. (a) Given a dataset with the following points: $(2,3), (3,3), (6,8), (8,8)$, apply one iteration of the k-means algorithm with $k = 2$ and initial centroids $(3,4)$ and $(7,7)$. Calculate the new centroids after this iteration. Draw an approximate scatterplot of the data, mark the old and the new centroids, and show how centroids change their positions over the iteration. Show all your steps.

(b) Given the following similarity matrix S for a set of 4 points:

$$S = \begin{bmatrix} 0 & 1 & 0.1 & 0 \\ 1 & 0 & 0.1 & 0 \\ 0.1 & 0.1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Compute the unnormalized graph Laplacian L and find its eigenvalues. Use the smallest two eigenvalues to explain the initial clustering of the points.

(10+10 = 20)

5. Given the following training data:

Point	Coordinates	Label
A	$(2,3)$	$+1$
B	$(3,2)$	$+1$
C	$(6,8)$	-1
D	$(8,6)$	-1

(a) Derive the primal objective function and constraints for this dataset assuming a linear SVM without slack variables.

(b) Solve the constrained optimization problem by hand calculations and determine the equation of the decision boundary. Plot the data points approximately, mark the support vectors, and draw the decision boundary. Infer the class label of a test point $(4,2)$ from the decision boundary you derived. Show all your work.

(6+(7+4+3) = 20)

Data Point	F1	F2	CL
1	A	5	Positive
2	B	7	Positive
3	A	8	Negative
4	C	6	Negative
5	B	9	Positive

Table 1: Dataset for Q. 6

6. (a) Consider a dataset (refer to Table 1) with the two attributes F1 and F2. Given the dataset, calculate the following:
   (a) Prior probabilities of each class.
   (b) Conditional probabilities of each feature given each class.
   (c) Predict the class label for a new data point with features F1='B' and F2=8 using the Naive Bayes Classifier.

   (b) Given a simple Convolutional Neural Network with the following configuration:
   - **Input image:** 4x4 grayscale image
   - **Convolutional layer:** 1 filter (2x2), stride = 1, padding = 0
   - **ReLU activation function**
   - **Pooling layer:** Max pooling with 2x2 filter, stride = 2

   Input image:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 2 & 2 & 0 & 2 \\ 1 & 3 & 1 & 2 \end{bmatrix}$$

   Filter (2x2):

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

   (a) Compute the output dimensions after the convolutional layer.
   (b) Perform the convolution operation and apply the ReLU activation function.
   (c) Compute the output dimensions after the pooling layer.
   (d) Perform the max pooling operation.

((3+3+3)+(3+4+2+2)= 20)

Best of luck!